Implicit Differentiation and Product Rule
And Quotient Rule and Chain Rule...

It is possible to combine what we have learned about implicit differentiation with what we already know about product rule and quotient rule and chain rule to find the derivative of some pretty complicated relations. Remember these things are only relations, not necessarily functions. Recall the example of $x^2 + y^2 = 25$, the circle!

So lets try some problems! For problems 1 and 2 find $\frac{dy}{dx}$.

1) $x^2y^2 = 6y^3x^5$

2) $x^2y^3 + x^2y^4 = 2$

3) Find the equation of the tangent line to $x^3y^2 = 8$ at the point (8,-1).

4) Find $\frac{d^2y}{dx^2}$ if $x^2 + y^2 = 25$ at the point (-3,4).

5) If $3x^2 - 2xy + 3y = 1$, then when $x = 2$, $\frac{dy}{dx} =$

(A) -12 (B) -10 (C) $-\frac{10}{7}$ (D) 12 (E) 32